



# The physics and chemistry of beer foam: a review

Charles W. Bamforth<sup>1</sup>

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## Abstract

The assurance of a stable and appealing foam on beer requires an understanding and application of several physical and chemical principles and these are reviewed in this paper. In terms of physics, it is essential that the foam is produced efficiently, the presence of nucleation sites being important, with major determining factors being temperature and carbonation level. The principle driving force for foam instability is disproportionation, wherein bubble collapse is caused by the passage of carbon dioxide from smaller to larger bubbles. As nitrogen does not readily make this passage, this gas allows for much more stable foams. The stability of foam depends on the balance of foam-positive foaming components (polypeptides, hop bitter acids, metal ions, melanoidins) over foam-negative entities (ethanol, lipids, detergents). The principal foam-stabilizing proteins contributed by malted barley are Protein Z4 and Lipid Transfer Protein (LTP1), but the fragments of hordein produced by proteolysis in malting and perhaps mashing have a greater ability to enter into the foam but display less ability to stabilize the bubbles. They compete with Protein Z and LTP1 which display less foamability but greater foam stability. Carbohydrate complexed with protein may have a significant role to play, and this is potentially the reason for the benefits to foam afforded by wheat. Kinetic models analogous to those employed in the study of enzyme kinetics are useful in terms of interpreting foaming systems, as are model systems.

**Keywords** Foam · Disproportionation · Beer composition · Model systems · pH

## Introduction

This author first reviewed the complexities of beer foam almost forty years ago [1]. More recent surveys of the topic have been published [2, 3] and a practical guide to ensuring foam excellence was issued [4]. This latter document explores the tools available to interpret shortcomings in foam performance as well as providing a holistic elucidation of raw material and process impacts on foam stability. The present paper seeks to update the reader on the underpinning science of the issue, rather than explore the practical approaches to improving foam performance in trade. In addition to the book by Bamforth [4], this was addressed by Lusk [3] and in a recent article [5]. Equally, this review does not

address methods for measuring foam stability, which were summarized in [1, 5].

It is evident that the appearance of a stable and attractive foam on beer results in favorable customer reaction [6, 7]. Even drinkers who routinely consume beer direct from the bottle or can select those beers with superior foams when shown images [6]. There are differences internationally with regard to the quantity of stable foam that is expected on a beer, with at one extreme being mliko beers poured in the Czech Republic that are essentially only foam in a glass and the other being beers with little foam, e.g., on certain cask-conditioned beers served direct from the barrel in parts of England. It may even be the case that there are gender differences in the appeal of certain foam attributes, such as lacing [8].

There are diverse aspects to any discussion of the foaming properties of beer. We can divide the issues into:

*Formation*—the production of the foam.

*Stability*—the retention of the head. Beer differs from other carbonated beverages in that it tends to keep the foam once

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Contribution for the Special Issue: The chemistry behind malt and beer production—from raw material to product quality.

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✉ Charles W. Bamforth  
Charlie.Bamforth@sierranevada.com

<sup>1</sup> Sierra Nevada Brewing Company, Chico, CA 95928, USA

formed. The production of a foam is in contradiction to the force of surface tension, by which water seeks to occupy the lowest possible surface area. The countering of this force by surface-active molecules in the beer is the basis of head retention.

*Lacing (cling)*—the tendency of the foam to adhere to the inside of the glass as the beer is consumed. As will be described in this paper, lacing is closely related to—and dependent on—head retention.

*Appearance*—foams differ in their esthetic appeal in terms of aspects such as color and texture. Although beer is sometimes poured from a novelty perspective with foams of different colors and accepting that foams may also have a brownish tinge due to the concentration of colored materials in the foam, it is usually considered that beer foam should be white. The color of the beer per se is usually not a factor, the whiteness of the foam reflecting (literally) the scattering of light. The amount of research in this area is limited and will not be discussed further.

## Foam formation

Beers are supersaturated solutions of carbon dioxide. Even so, they tend not to spontaneously generate a foam on dispense [9, 10]. A nucleation site has to be present. This could be a particle, an imperfection (scratch, etc) on the drinking glass or a pre-formed micro bubble that has arisen through an agitation event. Those beers that do spontaneously foam and display what is referred to as gushing [3] contain within them a nucleation promoter, notably particles formed from the intensely hydrophobic peptide hydrophobin produced by molds such as *Fusarium* [11], or from oxidized and dimerized hop resins [12].

“Creaming” or “beading” refers to the release and rising of bubbles in beer, which is both an attractive feature but also one that replenishes the foam on top of the product.

The extent of foam formation is highly dependent on the beer temperature and the carbon dioxide content [13]. Gas solubility decreases as the temperature is raised so foam formation is increased. It was shown [13] that foam formation is described by

$$a_n^0 = 3.11C + 0.0962\gamma - 218\rho + 216,$$

where  $a_n^0$  = initial nucleation activity,  $\gamma$  = surface tension,  $\rho$  = density,  $C$  = carbon dioxide content (vol CO<sub>2</sub>/vol beer).

The size of the bubble that is generated is determined [14] from the equation

$$R_b = [3R_m\gamma/\rho g]^{1/3},$$

where  $R_b$  = radius of bubble (m),  $R_m$  = radius of nucleation site (m),  $\gamma$  = surface tension (mN m<sup>-1</sup>),  $\rho$  = relative density of the beer (kg m<sup>-3</sup>),  $g$  = acceleration due to gravity (9.8 m s<sup>-2</sup>).

## Foam stability

### Physics

Various physical factors impact the stability of the foam once formed [14]. The first of these is drainage: any factor that slows the rate by which beer drains from the foam will increase its stability.

The rate of drainage is given by the equation [15]

$$Q = \frac{2\rho g q \delta}{3\eta},$$

where  $Q$  = flow rate (m<sup>3</sup> s<sup>-1</sup>),  $\eta$  = viscosity of film liquid (Pa s),  $\rho$  = density (kg m<sup>-3</sup>),  $q$  = length of Plateau border (m),  $g$  = acceleration due to gravity,  $\delta$  = thickness of film (m).

However, the most significant physical force leading to foam collapse is disproportionation, wherein gas passes from small bubbles to large ones, ultimately diminishing the total number of bubbles and rendering them larger on average. It is much more significant than coalescence, except when significant lipid contamination of beer occurs, for examples with unclean drinking glasses [16]. Disproportionation is summarized in the DeVries equation [16]

$$r_o^2 - r_t^2 = \frac{4RTDS\gamma t}{P\theta},$$

where  $r_o$  is bubble radius at the start,  $r_t$  is the bubble radius at time  $t$  (s),  $R$  is the gas constant (8.3 J K<sup>-1</sup> mol<sup>-1</sup>),  $T$  is absolute temperature (°K),  $D$  is the gas diffusion coefficient (m<sup>2</sup> s<sup>-1</sup>),  $S$  is the solubility of the gas (mol m<sup>-3</sup> Pa<sup>-1</sup>),  $\gamma$  is the surface tension (mN m<sup>-1</sup>),  $P$  is atmospheric pressure (Pa),  $\theta$  is the film thickness between bubbles (m).

The equation illustrates why the very insoluble gas nitrogen has a huge impact in enhancing foam stability, meaning that there is far less passage of molecules between the bubbles than occurs with carbon dioxide. Also significant is the film thickness, such that anything that retards the rate at which liquid drains between bubbles will enhance foam stability. Inter alia, drainage is less impeded when flowing around larger bubbles. This refocuses our attention on factors impacting bubble size and drainage (see earlier).

### Chemistry

The net foam stability observed in beer reflects the balance between the levels of those substances that promote head

retention and the concentration of inhibitors of foam. Thus, even if there are significant levels of foam-destabilizing molecules, they will be less able to exert their impact if there is an abundance of substances that interact to stabilize the foam.

The main endogenous foam stabilizers are the polypeptides and bitter acids, supported by melanoidins and metal ions. Principle amongst the foam-negative entities are ethanol and lipids. Thus, as an example, even a barley wine which may contain perhaps 10% alcohol by volume will still display a good foam, reflecting the customary high protein content from malt, relatively high bitterness and presence of melanoidins.

One of the challenges in interpreting the literature on foam is that diverse methods are available to quantify foam stability and lacing. Although there are several reports showing some degree of correlation between methods, this is not uniformly the case. Of equal concern is that it has

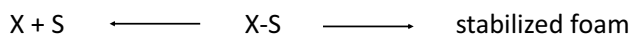
been shown that beer components may be shown to have a positive effect on foam stability using some methods, but not with others. For example, methods based on beer drainage over-emphasize factors that increase beer viscosity (see later).

### Polypeptides and foam

For a protein to function as a foam stabilizer, it must be readily adsorbed on the bubble interface but also be able to adapt, e.g., by unfolding into the most effective configuration, be mobile to overcome thinning in the surface and also favorably interact with other molecules, notably the iso- $\alpha$ -acids. The polypeptide must form a strong, flexible and cohesive film to reduce gas permeability and thereby inhibit disproportionation [17, 18].

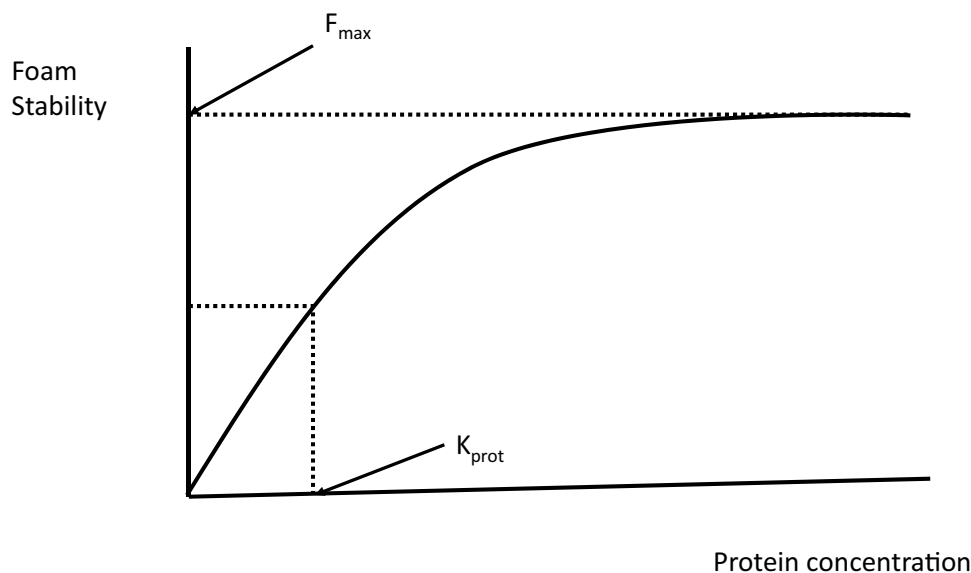
The foaming properties of polypeptides can be interpreted in a fashion analogous to that of the kinetics of enzymes (Fig. 1) [19, 20]. There is an initial reversible interaction of polypeptides (and other foam-stabilizing entities such as iso- $\alpha$ -acids) with the bubble interface, followed by an essentially irreversible transition (which would include molecular rearrangement and interactions between molecules) that results in the stabilization of that foam. Thus, just as an enzyme kineticist derives values for substrate affinity ( $K_m$ ) and maximum reaction rate ( $V_{max}$ ) using double-reciprocal plots of reaction rate versus substrate concentration, so can the foam scientist use equivalent plots of reciprocal foam stability against reciprocal polypeptide concentration to derive values for the affinity of a given polypeptide for the bubble ( $K_{prot}$ ) and the theoretical maximum foam stability ( $F_{max}$ ) that could be achieved for that polypeptide (Fig. 2a, b).

Using this approach, it was shown that the polypeptides derived from barley, subjected as they are to enzymatic



**Fig. 1** The rationale for the hyperbolic plot observed for an enzyme-catalyzed reaction is that there is a reversible interaction between the surface of the enzyme and the substrate, but at a sufficiently high substrate concentration all of the available sites on the enzyme are filled, and the enzyme is operating at maximum rate, which is determined by the rate at which the complex breaks down to reform the enzyme and release product). The analogous situation for foam would be if we consider the bubble surface (X) as being the equivalent of the enzyme surface, there being a reversible entry and departure of the surfactant molecules (S) to it. Changes can occur in the bubble (e.g., deformation of a polypeptide, interaction of the polypeptide with other components such as bitter acids) that leads to a stabilized surface. The hypothesis says that once all of the available space for the surfactant is occupied on the bubble (i.e., it is saturated) then the foam stability for a given polypeptide is maximized

**Fig. 2** The relationship between measured foam stability and polypeptide concentration mirrors that of the impact of substrate concentration in an enzyme-catalyzed reaction



hydrolysis during malting and to a lesser extent mashing, differ significantly in their affinity for the bubble interface and their subsequent ability to stabilize foam [21, 22]. Basically, the peptides derived by the enzymatic hydrolysis of hordein have superior ability to enter into foam (low  $K_{\text{prot}}$ ) but display less good foam stability ( $F_{\text{max}}$ ) than the albumin proteins that have attracted most attention as potential key players in determining the stability of beer foam (Table 1). It was noted that hydrolysis of the albumin fraction of barley proteins increase its  $F_{\text{max}}$  but lowered its affinity for the bubble interface [19] (Fig. 3).

Thus, in beer the net foam stability will not be determined by the absolute level of any individual protein species championed for its foaming properties, but rather by the relative levels of the albumin- and hordein-derived polypeptides. This explains the decrease in foam stability claimed to occur in beers produced from overly modified malts, which will contain increased quantities of hordein degradation products [23].

Nonetheless, the albumins Lipid Transfer Protein (LTP1) and Protein Z remain the most intensively studied polypeptides with regard to their foaming properties [24]. This reflects their relative abundance in beer, in part due to their resistance to protein digestion during malting and brewing [2]. Bech [25] showed that LTP1 isolated from barley, malt or sweet wort does not display especially good stabilizing properties, but these are greatly improved by the expedient of boiling the wort. This is almost certainly a reflection of the denaturation of the protein, leading to an increased exposure of the hydrophobic interior of the proteins, it being recognized that foam stability is greater for

**Table 1** Foam parameters

| Protein            | $K_{\text{prot}}$ (mg/mL) | $F_{\text{max}}$ (cm) |
|--------------------|---------------------------|-----------------------|
| Hydrolyzed hordein | 0.67                      | 16                    |
| Albumin            | 1.4                       | 46                    |
| Hydrolyzed albumin | 3.0                       | 89                    |

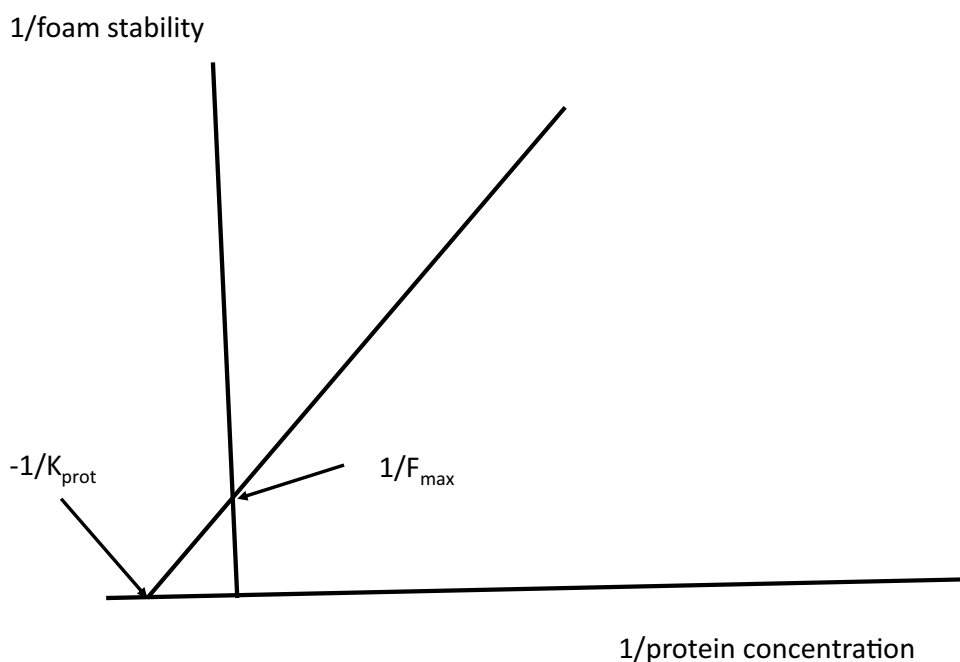
Foam was assessed by a shaking test in which the depth of foam (cm) surviving a stand of 30 min is measured. Data from [19]

relatively hydrophobic polypeptides [26, 27]. Van Nierop et al. [28] suggest that a major role for LTP1 in enhancing beer foam is via its role in binding foam-negative lipids and they suggest that the modifications to the polypeptide occurring during wort boiling lessen this ability. Commercial trials designed to lessen the extent of boiling do show improved foam stability if boil times are reduced [29].

Vaag et al. [30] later reported a third significant barley-derived foaming polypeptide, which they refer to as the 17 kDa protein. This polypeptide also seems to be transformed during wort production into a form that tends to concentrate more in foam.

The Protein Z4 component correlates with improved foam performance, but the converse was observed for protein Z7 [31]. Protein Z and an  $\alpha$ -amylase inhibitor correlate positively with foam; whereas, yeast-derived thioredoxin appeared to be foam negative [33, 33]. Niu et al. [34] say that protein Z is the most important protein for foam and question the opinion that it is the most hydrophobic polypeptides that are most foam stabilizing. They add that glycation of protein Z enhances its foaming properties [35]. By

**Fig. 3** The use of a double-reciprocal plot to measure maximum foam stability ( $F_{\text{max}}$ ) and the affinity of a polypeptide for the foam ( $K_{\text{prot}}$ )



contrast, others report that Protein Z is the less important of these polypeptides [36].

It seems that there are two key protein fractions responsible for beer foaming: LTP1 and a higher molecular weight entity that embraces Protein Z but an even greater proportion of carbohydrate [37]. It is tempting to speculate that the latter fraction is of the type hypothesized by Bishop [38], who suggested that during the kilning and roasting of malted grains there is a melding of protein and polysaccharide through Maillard-type glycation reactions to produce materials with especially good foaming properties. It is likely that the hydrophobic polypeptide entity is responsible for bringing the species into the bubble interface and that the polysaccharide component serves to slow liquid drainage by increasing the localized viscosity in the beer between the bubbles. Petry-Podgorska et al. [39] characterized polypeptides modified by glycation using a combination of two-dimensional high performance liquid chromatography and matrix assisted laser desorption/ionization time of flight mass spectrometry (MALDI-TOF/TOF MS). Carbohydrate moieties attached to polypeptides decrease the extent of loss of foaming polypeptides such as LTP1 through the brewing process [40].

Wheat proteins have superior foam-stabilizing abilities to those from barley [41] and it is further claimed that arabinoxylan is concentrated in beer foam and it is suggested that this is important in the foam quality of wheat beers [42]. Others suggest that the superior foaming performance of such beers relates to their higher degree of carbonation [43]. Continuing the theme of the importance of polypeptide–carbohydrate complexes, it has been shown that yeast wall mannoprotein has good foam-stabilizing potential [44].

### Bitter acids and foam stability

The bitter acids derived from hops interact with the foaming polypeptides via hydrophobic and ionic interactions, with the added involvement of divalent metal cations [45]. The interactions lead to an increase in the surface viscosity in the bubble interface and this thixotropic effect (“solidification”) renders the foam able to adhere to the glass surface (lacing or cling). The isohumulones are more concentrated in foam than are the isocohumulones and are more foam positive [46, 47]. Although there is a slightly greater partitioning of trans isomers into foam, there does not seem to be a significant difference between cis and trans isomers in their ability to stabilize foam [48]. A range of minor hop resin components may also have a role to play [49].

It seems that it is the interaction between isohumulone and protein Z that is particularly relevant for bubble surface changes that stabilize foam [50, 51].

Reduced iso- $\alpha$ -acids introduced to protect beer from light deterioration, give particularly stable foams included enhanced lacing, but these are frequently deemed esthetically unappealing [2].

There is little understanding of the contribution of different metal ions in terms of helping to stabilize foam [45], although Lu et al. [52] report the ability of the bitter substances to self-associate with calcium, potassium, magnesium and sodium (all abundant in beer) increasing aggregation and surface adsorption. A clear candidate for potential addition to beer for the improvement of foam is zinc [8].

### Other contributors to foam stability

It has been argued that increased beer viscosity due to residual  $\beta$ -glucans and arabinoxylans leads to improved foam performance [53] but this is probably an artifact due to the use of methods for assessing foam stability that are based on liquid drainage, which exaggerates the importance of viscosity [54]. Furthermore, the increase in viscosity which might be argued would increase foam stability would be problematic in regard to solid–liquid separation processes in the brewery [55]. Neugrodda et al. [56] suggest that there is a benefit to foam from  $\beta$ -glucan in beers that have not been treated with exogenous enzymes, but the approach taken is to correlate measurements of foam stability with those of the polysaccharide levels measured in a range of beers. The problem with this approach is that there are very likely indirect correlations. Thus, it is equally likely that the factor actually benefitting foam is something that is correlated with the level of  $\beta$ -glucan, for example lower levels of hydrolyzed hordein in less well-modified malt. In the same vein, these authors in common with others (e.g., [53]) suggest that polyphenols have a direct positive impact on foam, but again this is more likely associated with other factors with which the polyphenol level correlated, e.g., bitter acids. A further example is Chen et al. [57], who suggest that malt oligosaccharides correlate with foam stability. This correlatory approach to foam stability is problematic.

One relevant impact of polyphenols may be their role in removing foam-promoting polypeptides during the brewing process [58].

Factors that enhance the surface viscosity (i.e., that in the foam per se), would, however, be expected to improve foam stability. This would be much more likely be due to complexes of polypeptides and polysaccharides as was originally proposed by Bishop [38], c.f. [37]. There are claims for a benefit to foam stability by the use of various gums [3], though the mechanism by which they would do this is unclear. There is a long history of using propylene glycol alginate as a stabilizer to benefit foam, the mechanism being



through the cross-linking of foaming polypeptides in the bubble interface [59].

Lusk et al. [60] claim a significant role for melanoidins in foam stability. Meanwhile there appears to be a rather different extremely foam-active species generated in malts (e.g., black malt) that are intensely heated [61]. This entity appears to be a complex of pyridyl products of roasting and small numbers of amino acids.

### Foam inhibitors

The primary foam-negative material found in beer is ethanol [62]. Although relatively low concentrations (1–3%, by volume) do promote the ability to form foam and may slightly benefit head retention as measured using certain methods, the more ethanol in beer then the greater the interference with foam stability. It is argued that ethanol inhibits foam in a manner similar to lipid by interfering with the ability of the foam-stabilizing entities of beer associating to produce a stable foam matrix in the interface [62].

The ethanol specification in beer is, of course, not a variable for any product. In terms of addressing foam-negative materials, therefore, there is more scope for attention to the next major category of inhibitors, the lipids [63]. These may be classified into endogenous lipids (those entering from raw materials and yeast during the process) and exogenous lipids (those introduced from external source in processing and into the final product).

Most of the lipid originating in the grist is adsorbed to spent grains and trub materials generated in the brewhouse. However, levels capable of inhibiting foam will be present in finished beer, both long chain fatty acids and shorter chain fatty acids, the latter being inhibitory to lacing [64]. Of especial significance, however, seem to be oxidized fatty acids, produced either enzymatically through the action of lipoxygenase in malt production and mashing [65] or non-enzymatically. It has been suggested that these materials are especially damaging to foam as they survive into beer more than the unoxidized lipids that will be consumed by yeast [2]. It has been shown that such materials are developed in significant quantities during the production of certain specialty malts [66, 67].

Lipids and detergents introduced when beer is dispensed at the bar are proportionately more deleterious to foam than are those introduced in the process [4].

The basic amino acids arginine, lysine and histidine have been shown to inhibit lacing [68].

### Model systems

It will be realized that a multiplicity of factors influences the stability of foam. One useful approach to breaking down the complexity of beer foaming has been to take model

approaches, for example [69]. It has been stressed that relevant models should also include foam-negative entities [70].

Bamforth and Kanauchi [71] examined the influence of diverse variables in beer by examining model mixtures of isolated barley polypeptides and iso- $\alpha$ -acids. Amongst the observations were that ethanol profoundly inhibited the foam stability of solutions of albumin proteins whereas that of the hordein-derived polypeptides was increased. In that hordein competes with albumins for space in the bubble interface (see earlier), this observation lends further strength to the argument that foam stability is determined by the relative proportions of the different polypeptides and not the absolute level of a specific type of polypeptide. It was also shown that zinc is much more effective in promoting the foam stability of albumins than hordein-derived polypeptides.

One of the beer variables that is still not fully understood in respect of impact on foam is pH. Melm and colleagues [72] claimed that foam stability was better at lower pH values in the typical range for most beers (4.0–4.5). In model work, however, it was found that for both albumins and hordeins there was an increase in foam stability as pH was raised over the range 3.8–4.6 [71]. This concurs with the observation of Siebert [36] with his model system involving ovalbumin. However, it has been shown that adjustment of pH downwards in beers per se tends to lead to an increase in foam stability for pale ales and lagers, though in the latter instance there is a decrease in foam stability as the pH is progressively lowered below 4 [73]. Clearly, there is much yet to understand about the impact of pH on foam stability.

## Conclusions

The achievement of an excellent foam on beer demands attention to several physical and chemical factors. Foam formation demands consideration of nucleation sites and the size of the bubbles generated at such sites, as the most stable and appealing foams demand a uniform population of small bubbles. While nitrogen gas can greatly improve foam stability by lessening the extent of disproportionation, it can have undesirable impacts on flavor, including an increased “smoothness” and decreased “prickle”, as well as a lowering of hop-derived aroma [74]. Stable foams are inherently dependent on the presence of surface-active substances that counter the force of surface tension and the net foam stability observed is a balance between the levels of positives and negatives. Of the latter, the most significant is ethanol, but the level of this molecule in beer is of course an unmovable feature for a given beer. The other main foam negatives are lipids and detergents, especially those present in a glass at the point of dispense. Lipids can emerge in beer from raw materials and of particular significance are oxidized fatty acids generated either in malt production or mashing. The

principal backbone material for foam in beer is polypeptide. Again, there is a balance between those proteins with strong foam-stabilizing ability (Protein Z, LTP1 and the 17 kDa protein) and those that have profound ability to enter into foams but that are less adept at stabilizing foams (proteolysis products from the prolamin storage proteins). There is an interaction between proteins and bitter acids in the bubble interface, seemingly most profoundly with protein Z. The interaction is promoted by divalent metal ions and the resultant complexes display an increased surface viscosity and resultant ability to lace the glass. It is recommended that future fundamental research on beer foam takes the approach of building up ever more comprehensive model systems. Existing studies of this type have shown a variance in behavior as compared to beer per se. For example, with isolated proteins there is an increase foam stability as the pH is raised in the range 4–4.5, whereas the opposite is the case with beers per se. It has been indicated that realistic models must include fatty acids, but equally it could be argued that other beer components reported to be relevant for foaming, e.g., melanoidins, free amino acids, cations should also be included. Of all the factors in beer that are routinely measured, perhaps it is indeed pH that has been the one that has attracted least attention in respect of foam performance. This needs rectification as part of the model system approach. Although much progress has been made over the decades in understanding the physics and chemistry of foam, there remain several incompletely understood aspects, notably in relation to chemistry. Further exploration needs to employ tools that are not going to generate conclusions dependent on the procedure used to quantify the foam. On that basis, approaches must be instrumentally focused on the nature of the molecules that are present in foams and of the interactions that are occurring between them. Only then will it be possible to state with more certainty the relative significance of the diverse polypeptides found in beer, precisely how they interact with the bitter acids and metal ions, and to what extent there is or is not a significant role played by substances such as Maillard reaction products, polysaccharides, and polyphenols.

## Declarations

**Conflict of interest** The author declares that he has no conflict of interest.

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